

### Problem Statement 1

1. Load the given iris data set into data frame.
2. Add column names to data frame. (if column names are not present)
3. Display first and last five rows of data frame.
4. Display initial statistics of data frame.
5. Display variable descriptions
6. Check data types of columns in data frame and if variables are not in the correct data type, apply proper type conversions.
7. Apply Data Normalization and Standardization.
8. Perform label encoding and convert categorical variable into quantitative variables. Use any label encoding technique and apply it on 'Species' column of data frame.

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### Problem Statement 2

#### Part A:

1. Load the given academic performance data set into data frame
2. Check for missing values
3. filling missing values using mean, median and standard deviation of that column ( Note: use fillna())

#### Part B:

1. Load the data set demo.csv into data frame
2. Display the above data frame
3. Select the columns for boxplot and draw the boxplot
4. print the outliers for each column with reference to the box plot.
5. Detecting outliers for math\_score column using z-score normalization

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### Problem Statement 3

#### Part A:

1. Load the dataset eg. Mall\_Customers.csv into data frame.
2. Find mean, median, mode and standard deviation of all columns in data frame.
3. Provide summary statistics of income group by the age groups.
4. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable.

#### Part B:

1. Load the given iris data set into data frame.
2. Add column names to data frame. (if column names are not present)
3. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset

### Problem Statement 4:

Create a Linear Regression Model using Python/R to predict home prices using Boston Housing Dataset (<https://www.kaggle.com/c/boston-housing>). The Boston Housing dataset contains information about various houses in Boston through different parameters. There are 506 samples and 14 feature variables in this dataset.

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### Problem Statement 5:

1. Implement logistic regression using Python/R to perform classification on Social\_Network\_Ads.csv dataset.
2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.

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### Problem Statement 6:

1. Implement Simple Naïve Bayes classification algorithm using Python/R on iris.csv dataset.
2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset

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### Problem Statement 7:

#### Part A:

1. Extract Sample document and apply following document preprocessing methods:
  - a. Tokenization
  - b. POS Tagging
  - c. stop words removal
  - d. Stemming and Lemmatization.

#### Sample Text

"Tokenization is the first step in text analytics. The process of breaking down a text paragraph into smaller chunks such as words or sentences is called Tokenization."

#### Part B:

2. Create representation of document by calculating Term Frequency and Inverse Document Frequency.

#### Sample Document

documentA = 'Jupiter is the largest Planet'

documentB = 'Mars is the fourth planet from the

Sun'

### Problem Statement 8:

1. Use the inbuilt dataset 'titanic'. The dataset contains 891 rows and contains information about passengers who boarded the unfortunate Titanic ship. Use the Seaborn library to see if we can find any patterns in the data. (Draw Distribution Plot, Categorical Plots etc. and record your observations)
2. Write a code to check how the price of the ticket (column name: 'fare') for each passenger is distributed by plotting a histogram. (Draw histogram plot and record your observation)
3. Compare histograms of age for survivors vs non-survivors.
4. Use a bar plot to see survival counts grouped by embarkation port (embarked).
5. Generate correlation heatmap for numeric features

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### Problem Statement 9:

1. Use the inbuilt dataset 'titanic'. The dataset contains 891 rows and contains information about passengers who boarded the unfortunate Titanic ship. Use a bar plot to compare survival counts across passenger classes
2. Plot histograms of age for survivors vs non-survivors.
3. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not. (Column names : 'sex' and 'age')
4. Write observations on the inference from the above statistics.
5. Generate correlation heatmap for numeric features
6. Use a bar plot to see survival counts grouped by embarkation port (embarked)

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### Problem Statement 10:

1. Load the Iris flower dataset into a DataFrame.
2. List down the features and their types (e.g., numeric, nominal) available in the dataset.
3. Create a histogram for each feature in the dataset to illustrate the feature distributions.
4. Create a box plot for each feature in the dataset.
5. Compare distributions and identify outliers.
6. Use a pairplot to visualize relationships between features, colored by species.
7. Plot scatter plots between two numeric features to check relationship.
8. Generate a correlation heatmap for numeric features to identify strong/weak relationships.